

# Diana Halikias

## CONTACT

---

Warren Weaver Hall, Office 922  
Department of Mathematics, Courant Institute  
New York University

+1(240)-938-2639  
[diana.halikias@nyu.edu](mailto:diana.halikias@nyu.edu)  
[dihalikias.github.io](https://dihalikias.github.io)

## RESEARCH INTERESTS

---

Numerical analysis, randomized linear algebra, operator learning, matrix theory, approximation theory

## EDUCATION

---

- 2020 - 2025    Ph.D. in MATHEMATICS, **Cornell University**, Ithaca, NY  
Advisor: Alex Townsend  
Thesis: *Structured matrix recovery and approximation from matrix-vector products*  
NSF Graduate Research Fellow
- 2016 - 2020    B.A. (Intensive) in MATHEMATICS with distinction, **Yale University**, New Haven, CT  
Thesis: *Determinantal Processes: Discrete and Continuous Cases*, advised by Richard Kenyon

## EXPERIENCE

---

- Jan. 2026 - present    Courant Instructor, **Courant Institute, New York University**, New York, NY  
Faculty advisor: Chris Musco
- Fall 2025                Simons-Berkeley Postdoctoral Fellow, **Simons Institute for the Theory of Computing, UC Berkeley**, Berkeley, CA
- Summer 2024           Summer Research Associate, **Flatiron Institute**, New York, NY  
Research mentor: Lawrence Saul
- Summer 2023           Research Intern, **Lawrence Berkeley National Laboratory**  
Research mentors: Ben Erichson, Michael Mahoney, Dmitriy Morozov
- Summer 2019           Summer Researcher, **Weizmann Institute**, Rehovot, Israel  
Research mentor: Bo'az Klartag

## PUBLICATIONS AND PREPRINTS

---

- N. BOULLÉ, D. HALIKIAS, S. OTTO, AND A. TOWNSEND. *A zero-one law for one-shot system identification*. (2026) [arXiv:2607.15832](https://arxiv.org/abs/2607.15832)
- D. HALIKIAS, M.E. HOCHSTENBACH, AND A. TOWNSEND. *Transpose-free linear algebra*. (2026) [arXiv:2606.01335](https://arxiv.org/abs/2606.01335)
- M. F. SERRET, A. CORTINOVIS, Y. DONG, D. HALIKIAS, A. MA, F. MATTI, D. NEEDELL, K. J. PEARCE, E. REBROVA, D. SHUR, R. SMITH, H.-X. WANG, AND L. GRIGORI. *Attention Mechanisms Through the Lens of Numerical Methods: Approximation Methods and Alternative Formulations*. (2026) [arXiv:2604.01757](https://arxiv.org/abs/2604.01757)
- N. AMSEL, T. CHEN, F. DUMAN KELES, D. HALIKIAS, C. MUSCO, C. MUSCO, AND D. PERSSON. *Quasi-optimal hierarchically semi-separable matrix approximation*. SIAM J. Matrix Anal. Appl. 47, no. 2 (2026): 586-621. [doi:10.1137/25M176622X](https://doi.org/10.1137/25M176622X)
- N. AMSEL, T. CHEN, F. DUMAN KELES, D. HALIKIAS, C. MUSCO, AND C. MUSCO. *Fixed-sparsity matrix approximation from matrix-vector products*. SIAM J. Matrix Anal. Appl. 47, no. 2 (2026): 483-511. [doi:10.1137/25M1742710](https://doi.org/10.1137/25M1742710)
- T. CHEN, F. DUMAN KELES, D. HALIKIAS, C. MUSCO, C. MUSCO, AND D. PERSSON. *Near-optimal hierarchical matrix approximation from matrix-vector products*. Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 2656-2692 (2025). [doi:10.1137/1.9781611978322.87](https://doi.org/10.1137/1.9781611978322.87)

- N. BOULLÉ, D. HALIKIAS, S. OTTO, AND A. TOWNSEND. *Operator learning without the adjoint*. J. Mach. Learn. Res. 25, no. 364 (2024): 1-54. [JMLR:v25:24-0162](#)
- N. BOULLÉ, D. HALIKIAS, AND A. TOWNSEND. *Elliptic PDE learning is provably data-efficient*. PNAS 120, no. 39 (2023): e2303904120. [doi:10.1073/pnas.2303904120](#)
- D. HALIKIAS AND A. TOWNSEND. *Structured matrix recovery from matrix-vector products*. Numer. Linear Algebra Appl. 31, no. 1 (2024): e2531. [doi:10.1002/nla.2531](#)
- A. YU, C. BECQUEY, D. HALIKIAS, M. E. MALLORY, AND A. TOWNSEND. *Arbitrary depth universal approximation theorems for operator neural networks*. [arXiv:2109.11354](#).
- D. HALIKIAS, B. KLARTAG, AND B. SLOMKA. *Discrete variants of Brunn-Minkowski type inequalities*. Ann. Fac. Sci. Toulouse Math. (6), Vol. 30, no. 2, (2021), 267—279. [doi:10.5802/afst.1674](#)
- E. GELERT, D. HALIKIAS, C. KENNEY, AND N. MARSHALL. *A Cheeger inequality for graphs based on a reflection principle*. Involve 13 no. 3, (2020) 475–486. [doi:10.2140/involve.2020.13.475](#)

## INVITED TALKS

---

|                |                                                                                       |
|----------------|---------------------------------------------------------------------------------------|
| September 2026 | Rutgers/DIMACS Theory of Computing Seminar, Rutgers University                        |
| June 2026      | Randomized Numerical Linear Algebra workshop, EPFL, Lausanne                          |
| May 2026       | 27th Conference of the International Linear Algebra Society, Virginia Tech            |
| April 2026     | Applied and Computational Mathematics seminar, Temple University                      |
| April 2026     | Rising Stars in Computational and Data Sciences workshop, Santa Fe Institute          |
| March 2026     | IDeAS (Applied & Computational Math) seminar, Princeton University                    |
| March 2026     | Computational Mathematics and Scientific Computing Seminar, New York University       |
| October 2025   | Linear Systems and Eigenvalues workshop, Simons Institute for the Theory of Computing |
| June 2025      | The 30th Biennial Numerical Analysis Conference, University of Strathclyde            |
| June 2025      | Householder Symposium XXII, Cornell University                                        |
| March 2025     | Imperial-UCL Numerics Seminar                                                         |
| March 2025     | Computational Mathematics and Applications Seminar, Oxford University                 |
| January 2025   | Applied mathematics colloquium, Columbia University                                   |
| May 2024       | SIAM-Linear Algebra, Sorbonne University                                              |
| October 2023   | SIAM NY-NJ-PA section meeting, New Jersey Institute of Technology                     |
| August 2023    | ICIAM at Waseda University, Tokyo, Japan                                              |
| March 2023     | Olivetti Club, Cornell University                                                     |
| November 2022  | Olivetti Club, Cornell University                                                     |

## CONTRIBUTED TALKS AND POSTERS

---

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| February 2026 | ICERM Workshop on Randomized Linear Algebra, Providence, RI (poster)                                              |
| November 2024 | SIAM NY-NJ-PA section meeting, Rochester Institute of Technology                                                  |
| October 2024  | SIAM-Mathematics of Data Science, Atlanta, Georgia (poster)                                                       |
| July 2023     | Synergistic Interactions between Theory and Computation, MIT (poster)                                             |
| June 2023     | Modern Applied and Computational Analysis at ICERM, Brown University (poster)                                     |
| March 2022    | Latest trends and insights into matrix theory, iterative methods, and preconditioning, Temple University (poster) |

## TEACHING EXPERIENCE

---

### New York University

Spring 2026      **Instructor**, Math UA-185.007 (Probability and Statistics)

### Cornell University

Fall 2023      **Teaching assistant**, Math 2210 (Linear Algebra)

### Yale University

Spring 2020      **Teaching assistant**, Math 231 (Vector Calculus and Linear Algebra II)  
 Fall 2019      **Teaching assistant**, Math 230 (Vector Calculus and Linear Algebra I)  
 Spring 2019      **Teaching assistant**, Math 244 (Discrete Mathematics)  
 Fall 2018      **Teaching assistant**, Math 115 (Calculus II)

## RESEARCH FELLOWSHIPS AND AWARDS

---

- 2026 **AMS-Simons Travel Grant** (\$6,000)  
2026 **Rising Stars in Computational and Data Sciences**  
2024 **Cornell Graduate School Research Travel Grant** (\$1,300)  
2024 **Cornell Graduate School Conference Grant** (\$400)  
2024 **SIAM Student Travel Grant** (\$800)  
2023 **Cornell Graduate School Conference Grant** (\$700)  
2021 **NSF Graduate Research Fellow** (\$138,000)  
Awarded 5-year NSF-GRFP fellowship providing 3 years of financial support.  
2020 **Cornell Mathematics First Year Fellowship** (\$32,000)  
2019 **Yale-Weizmann Summer Science Fellowship**  
Awarded travel funding to conduct math research at the Weizmann Institute in Rehovot, Israel.

## SERVICE AND OUTREACH

---

- 2024 - 2025 **Cornell math department teaching seminar**, Organizer  
2020 - 2025 **Association for Women in Mathematics Mentorship Program**, Mentor  
Mentored three undergraduate women, provided support and guidance on early mathematical careers.  
2024 - 2025 **Olivetti seminar**, Organizer  
2022 - 2024 **Directed Reading Program**, Mentor  
Mentored two Cornell undergraduates in reading projects on spectral graph theory.  
2021 **Cornell Mathematics Department REU**, Mentor  
Mentored undergraduate research project in theoretical aspects of deep learning.  
2021 - 2023 **Graduate Mentorship Program**, Mentor  
Mentored incoming graduate students at Cornell on transitioning to graduate life.  
2021 - 2023 **Little Math Circle**, Leader  
Lead and teach weekly creative math circle for elementary school students in the Ithaca area.  
2020 - 2021 **Cornell Association for Women in Mathematics Chapter**, Officer  
Taught  $\text{\LaTeX}$ workshop for Cornell women in STEM.  
2017 - 2020 **Yale Association for Women in Mathematics Student Chapter**, President

## ACTIVITIES

---

- 2025 **IPAM Research Collaboration Workshop: Randomized Numerical Linear Algebra (RNLA)**  
Participated in research workshop on low-rank approximation and transformers.  
2022 **MSRI Summer Graduate School on Mathematics of Machine Learning (Courant Institute)**  
Nominated by Cornell's math department to attend funded MSRI graduate school on machine learning theory.  
2018 **Summer Undergraduate Research in Mathematics at Yale (SUMRY)**  
Awarded funding to participate in Yale's Research Experience for Undergraduates in mathematics.

## TECHNICAL SKILLS AND LANGUAGES

---

PROGRAMMING: Matlab, Python,  $\text{\LaTeX}$   
LANGUAGES: English, French, Modern Greek, Chinese (Mandarin)

## MISCELLANEOUS

---

- Host of weekly Cornell graduate student trivia night at the Big Red Barn (2021-2024)
- Awarded second place Montaigne Prize, Yale University Department of French (2020)
- Former student employee in the Numismatics Department of the Yale University Art Gallery (2017-2020)
- Silver Medalist in piano performance, AADGT International Competition; performed twice at Carnegie Hall (2014)